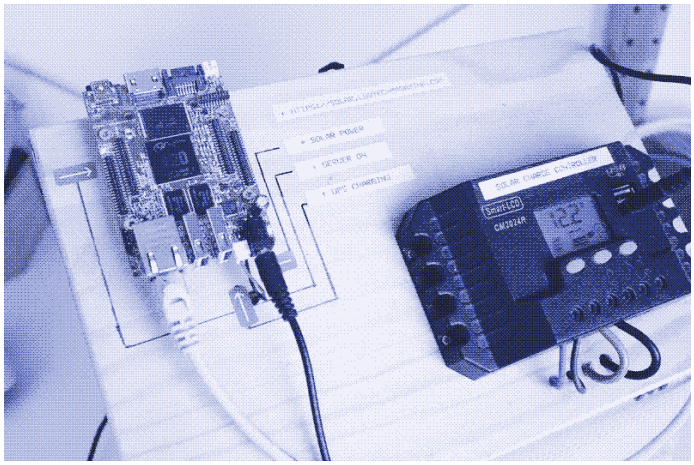
LOW←TECH MAGAZINE
This is a solar-powered website, which means it sometimes goes offline | visit at: https://solar.lowtechmagazine.com

How to Build a Low-tech Website?

Our new website is designed to radically reduce the energy use associated with accessing our content.

September 24, 2018 Written by Kris De Decker Marie Otsuka Roel Roscam Abbing Translations fr de nl es pt pl



First prototype of the solar powered server that runs the new website. The solar charge controller (on the right) is powering the server (on the left) through a USB-cable.

Low-tech Magazine was born in 2007 and has seen minimal changes ever since. Because a website redesign was long overdue — and because we try to practice what we preach — we decided to build a low-tech, self-hosted, and solar-powered version of Low-tech Magazine. The new blog is designed to radically reduce the energy use associated with accessing our content.

Why a Low-tech Website?

We were told that the Internet would "dematerialise" society and decrease energy use (https://www.bcg.com/publications/2012/energy-environment-technology-industries-smarter-2020-role-ict-driving-sustainable-future.aspx) . Contrary to this projection, it has become a large and rapidly growing consumer of energy itself (https://solar.lowtechmagazine.com/2015/10/why-we-need-a-speed-limit-for-the-internet/) .

In order to offset the negative consequences associated with high energy consumption, renewable energy has been proposed as a means to lower emissions from powering data centers. For example, Greenpeace's yearly ClickClean report (http://www.greenpeace.org/usa/global-warming/click-clean/) ranks major Internet companies based on their use of renewable power sources.

However, running data centers on renewable power sources is not enough to address the growing energy use of the Internet. To start with, the Internet already uses three times more energy than all wind and solar power sources worldwide can provide. Furthermore, manufacturing, and regularly replacing, renewable power plants also requires energy (https://solar.lowtechmagazine.com/2015/04/how-sustainable-is-pv-solar-power/) , meaning that if data traffic keeps growing, so will the use of fossil fuels.

> Running data centers on renewable power sources is not enough to address the growing energy use of the Internet.

Finally, solar and wind power are not always available, which means that an Internet running on renewable power sources would require infrastructure for energy storage and/or transmission that is also dependent on fossil fuels for its manufacture and replacement (https://solar.lowtechmagazine.com/2017/09/how-not-to-run-a-modern-society-on-solar-and-wind-power-alone/) . Powering websites with renewable energy is not a bad idea, however the trend towards growing energy use must also be addressed.

To start with, content is becoming increasingly resource-intensive. This has a lot to do with the growing importance of video, but a similar trend can be observed among websites. The size of the average web page (https://httparchive.org/reports/page-weight) (defined as the average page size of the 500,000 most popular domains) increased from 0.45 megabytes (MB) in 2010 to 1.7 megabytes in June 2018. For mobile websites, the average "page weight" rose tenfold from 0.15 MB in 2011 to 1.6 MB in 2018. Using different measurement methods, other sources report average page sizes of up to 2.9 MB in 2018.

The growth in data traffic surpasses the advances in energy efficiency (https://www.researchgate.net/publication/224224694/download) (the energy required to transfer 1 megabyte of data over the Internet),

resulting in more and more energy use. “Heavier” or “larger” websites not only increase energy use in the network infrastructure, but they also shorten the lifetime of computers — larger websites require more powerful computers to access them. This means that more computers need to be manufactured, which is a very energy-intensive process (https://solar.lowtechmagazine.com/2009/06/the-monster-footprint-of-digital-technology/) .

> Being always online doesn’t combine well with renewable energy sources such as wind and solar power, which are not always available.

A second reason for growing Internet energy consumption is that we spend more and more time on-line. Before the arrival of portable computing devices and wireless network access, we were only connected to the network when we had access to a desktop computer in the office, at home, or in the library. We now live in a world in which no matter where we are, we are always on-line, including, at times, via more than one device simultaneously.

“Always-on” Internet access is accompanied by a cloud computing model – allowing more energy efficient user devices at the expense of increased energy use in data centers. Increasingly, activities that could perfectly happen off-line – such as writing a document, filling in a spreadsheet, or storing data – are now requiring continuous network access. This does not combine well with renewable energy sources such as wind and solar power, which are not always available.

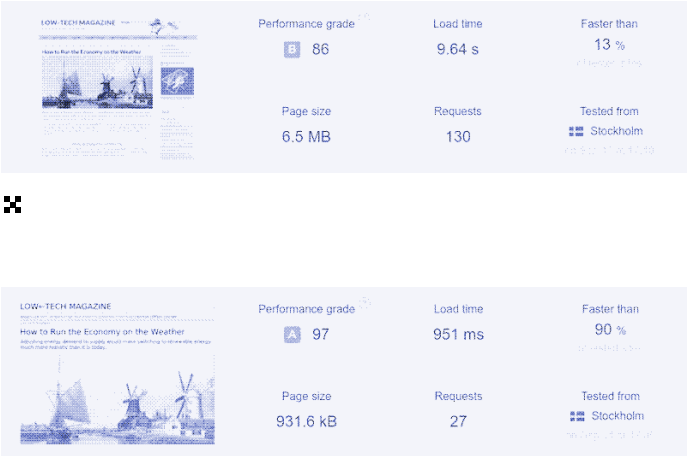
Low-tech Web Design

Our new web design addresses both these issues. Thanks to a low-tech web design, we managed to decrease the average page size of the blog by a factor of five compared to the old design – all while making the website visually more attractive (and mobile-friendly). Secondly, our new website runs 100% on solar power, not just in words, but in reality: it has its own energy storage and will go off-line during longer periods of cloudy weather.

The Internet is not an autonomous being. Its growing energy use is the consequence of actual decisions (http://tonsky.me/blog/disenchantment/) made by software developers, web designers, marketing departments, publishers and internet users. With a lightweight, off-the-grid solar-powered website, we want to show that other decisions can be made.

> With 36 of roughly 100 articles now online, the average page weight on the solar powered website is roughly five times below that of the previous design.

To start with, the new website design reverses the trend towards increasingly larger page sizes. With 36 of roughly 100 articles now online, the average page weight on the solar powered website is 0.77 MB — roughly five times below that of the previous design, and less than half the average page size of the 500,000 most popular blogs in June 2018.



A web page speed test from the old and the new Low-tech Magazine. Page size has decreased more than sixfold, number of requests has decreased fivefold, and download speed has increased tenfold. Note that we did not design the website for speed, but for low energy use. It would be faster still if the server would be placed in a data center and/or in a more central location in the Internet infrastructure. Source: Pingdom.

Below are some of the design decisions we made to reduce energy use. We have published a separate document that focuses on the front-end efforts (https://github.com/lowtechmag/solar/wiki/Solar-Web-Design) , and one that focuses on the back-end (https://homebrewserver.club/low-tech-website-howto.html#software) . We have also released the source code for our website design (https://github.com/lowtechmag/solar) .

Static Site

One of the fundamental choices we made was to build a static website. Most of today’s websites use server side programming languages that generate the website on the fly by querying a database. This means that every time someone visits a web page, it is generated on demand.

On the other hand, a static website is generated once and exists as a simple set of documents on the server’s hard disk (https://varia.zone/en/what-a-website-can-be.html) . It’s always there –

not just when someone visits the page. Static websites are thus based on file storage whereas dynamic websites depend on recurrent computation. Static websites consequently require less processing power and thus less energy.

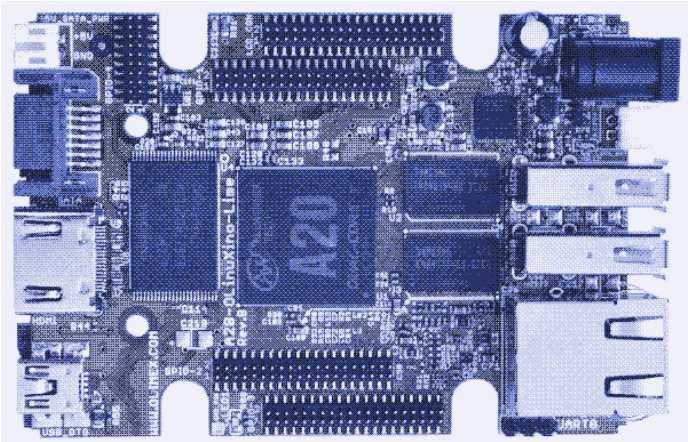
The choice for a static site enables the possibility of serving the site in an economic manner from our home office in Barcelona. Doing the same with a database-driven website would be nearly impossible, because it would require too much energy. It would also be a big security risk. Although a web server with a static site can be hacked, there are significantly less attack routes and the damage is more easily repaired.

Dithered Images

The main challenge was to reduce page size without making the website less attractive. Because images take up most of the bandwidth, it would be easy to obtain very small page sizes and lower energy use by eliminating images, reducing their number, or making them much smaller. However, visuals are an important part of Low-tech Magazine's appeal, and the website would not be the same without them.

> By dithering, we can make images ten times less resource-intensive, even though they are displayed much larger than on the old website.

Instead, we chose to apply an obsolete image compression technique called "dithering". The number of colours in an image, combined with its file format and resolution, contributes to the size of an image. Thus, instead of using full-colour high-resolution images, we chose to convert all images to black and white, with four levels of grey in-between.



A dithered image of our server.

These black-and-white images are then coloured according to the pertaining content category via the browser's native image manipulation capacities. Compressed through this dithering plugin, images featured in the articles add much less load to the content: compared to the old website, the images are roughly ten times less resource-intensive.

Default typeface / No logo

All resources loaded, including typefaces and logos, are an additional request to the server, requiring storage space and energy use. Therefore, our new website does not load a custom typeface and removes the font-family declaration, meaning that visitors will see the default typeface of their browser.

We use a similar approach for the logo. In fact, Low-tech Magazine never had a real logo, just a banner image of a spear held as a low-tech weapon against prevailing high-tech claims.

Instead of a designed logotype, which would require the production and distribution of custom typefaces and imagery, Low-tech Magazine's new identity consists of a single typographic move: to use the left-facing arrow in place of the hypen in the blog's name: LOW←TECH MAGAZINE.

No Third-Party Tracking, No Advertising Services, No Cookies

Web analysis software such as Google Analytics records what happens on a website — which pages are most viewed, where visitors come from, and so on. These services are popular because few people host their own website. However, exchanging these data between the server and the computer of the webmaster generates extra data traffic and thus energy use.

With a self-hosted server, we can make and view these measurements on the same machine: every web server generates logs of what happens on the computer. These (anonymous) logs are only viewed by us and are not used to profile visitors.

> With a self-hosted server, there's no need for third-party tracking and cookies.

Low-tech Magazine has been running Google Adsense advertisements since the beginning in 2007. Although these are an important financial resource to maintain the blog, they have two important downsides. The first is energy use: advertising services raise data traffic and thus energy use.

Secondly, Google collects information from the blog's visitors, which forces us to craft extensive privacy

statements and cookie warnings — which also consume data, and annoy visitors. Therefore, we replace Adsense by other financing options (read more below). We use no cookies at all.

How often will the website be off-line?

Quite a few web hosting companies claim that their servers are running on renewable energy. However, even when they actually generate solar power on-site, and do not merely "offset" fossil fuel power use by planting trees or the like, their websites are always on-line.

This means that either they have a giant battery storage system on-site (which makes their power system unsustainable), or that they are relying on grid power when there is a shortage of solar power (which means that they do not really run on 100% solar power).

The 50W solar PV panel. On top of it is a 10W panel powering a lighting system.

In contrast, this website runs on an off-the-grid solar power system with its own energy storage, and will go off-line during longer periods of cloudy weather. Less than 100% reliability is essential for the sustainability of an off-the-grid solar system, because above a certain threshold the fossil fuel energy used for producing and replacing the batteries is higher than the fossil fuel energy saved by the solar panels.

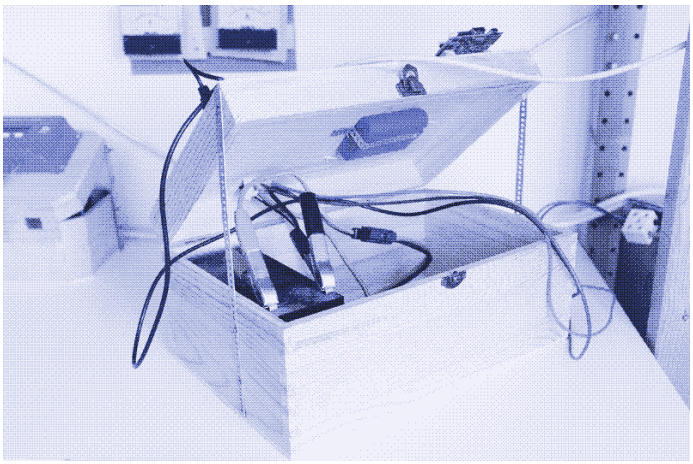
How often the website will be off-line remains to be seen. The web server is now powered by a new 50 Wp solar panel and a two year old 12V 7Ah lead-acid battery. Because the solar panel is shaded during the morning, it receives direct sunlight for only 4 to 6 hours per day. Under optimal conditions, the solar panel thus generates 6 hours x 50 watt = 300 Wh of electricity.

The web server uses between 1 and 2.5 watts of power (depending on the number of visitors), meaning that it requires between 24 Wh and 60 Wh of electricity per day. Under optimal conditions, we should thus have sufficient energy to keep the web server running for 24 hours per day. Excess energy production can be used for household applications.

> We expect to keep the website on-line during one or two days of bad weather, after which it will go off-line.

However, during cloudy days, especially in winter, daily energy production could be as low as 4 hours x 10 watts = 40 watt-hours per day, while the server requires beteen 24 and 60 Wh per day. The battery storage is roughly 40 Wh, taking into account 30% of charging and 33% depth-or-discharge (the solar charge controller shuts the system down when battery voltage drops to 12V).

Consequently, the solar powered server will remain on-line during one or two days of bad weather, but not for longer. However, these are estimations, and we may add a second 7 Ah battery in autumn if this is necessary. We aim for an "uptime" of 90%, meaning that the website will be off-line for an average of 35 days per year.



First prototype with lead-acid battery (12V 7Ah) on the left, and Li-Po UPS battery (3,7V 6600mA) on the right. The lead-acid battery provides the bulk of the energy storage, while the Li-Po battery allows the server to shut down without damaging the hardware (it will be replaced by a much smaller Li-Po battery).

When is the best time to visit?

The accessibility of this website depends on the weather in Barcelona, Spain, where the solar-powered web server is

located. To help visitors "plan" their visits to Low-tech Magazine, we provide them with several clues.

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A battery meter provides crucial information because it may tell the visitor that the blog is about to go down – or that it's "safe" to read it. The design features a background colour that indicates the capacity of the solar-charged battery that powers the website server. A decreasing height indicates that night has fallen or that the weather is bad.

In addition to the battery level, other information about the website server is visible with a statistics dashboard. This includes contextual information of the server's location: time, current sky conditions, upcoming forecast, and the duration since the server last shut down due to insufficient power.

To access Low-tech Magazine no matter the weather, we have several offline reading options available (/offline-reading.html) .

Hardware and Software

We wrote two extra articles with more in-depth technical information: How to build a low-tech website: software and hardware (https://homebrewserver.club/low-tech-website-howto.html) , which focuses on the back-end, and How to Build a Low-tech Website: Design Techniques and Process (https://github.com/lowtechmag/solar/wiki/Solar-Web-Design) , which focuses on the front-end.

SERVER: This website runs on an Olimex A20 computer (https://homebrewserver.club/low-tech-website-howto.html#server) . It has 2 Ghz of processing power, 1 GB of RAM, and 16 GB of storage. The server draws 1 - 2.5 watts of power.

SERVER SOFTWARE: The webserver runs Armbian Stretch, a Debian based operating system built around the SUNXI kernel. We wrote technical documentation for configuring the webserver (https://homebrewserver.club/low-tech-website-howto.html#configuring-the-webserver) .

DESIGN SOFTWARE: The website is built with Pelican (https://blog.getpelican.com/) , a static site generator. We have released the source code for 'solar', the Pelican theme we developed here (https://github.com/lowtechmag/solar) .

INTERNET CONNECTION. The server is connected to a 100 MBps fibre internet connection. Here's how we configured the router (https://homebrewserver.club/low-tech-website-howto.html#network) . For now, the router is powered by grid electricity and requires 10 watts of power. We are investigating how to replace the energy-hungry router with a more efficient one that can be solar-powered, too.

SOLAR PV SYSTEM. The server runs on a 50 Wp solar panel and one 12V 7Ah lead-acid battery. However, are still downsizing the system and are experimenting with different setups. The PV installation is managed by a 20A solar charge controller.

What happens to the old website?

The solar powered Low-tech Magazine is a work in progress. For now, the grid-powered Low-tech Magazine remains on-line. Readers will be encouraged to visit the solar powered website if it is available. What happens later, is not yet clear. There are several possibilities, but much will depend on the experience with the solar powered server.

Until we decide how to integrate the old and the new website, making and reading comments will only be possible on the grid-powered Low-tech Magazine, which is still hosted at TypePad. If you want to send a comment related to the solar powered web server itself, you can do so by sending an e-mail to solar (at) lowtechmagazine (dot) com. Your comment will be published at the bottom of this page.

Financing

We're hoping for people to support this project with a financial contribution. Advertising services, which have maintained Low-tech Magazine since its start in 2007, are not compatible with our lightweight web design. Therefore, we are searching for other ways to finance the website:

We offer print-on-demand copies of the blog (https://solar.lowtechmagazine.com/2019/03/low-tech-magazine-the-printed-website/) . These publications allow you to read Low-tech Magazine on paper, on the beach, in the sun, or whenever and where ever you want.

You can support us through through PayPal (https://www.paypal.me/lowtechmagazine) , Patreon (https://www.patreon.com/lowtechmagazine) and LiberaPay (https://liberapay.com/lowtechmagazine/) .

The solar powered website was made by Kris De Decker (http://www.krisdedecker.org) , Roel Roscam Abbing (https://roelof.info) , and Marie Otsuka (http://motsuka.com) . The printed website is made by Lauren Campbell (http://squishysystems.com) .

Related Articles

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How to Build a Low-tech Internet

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October 26, 2015

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We have launched the first volume in a new series of books opening up Low-tech Magazine's archive by theme.

August 18, 2023

Rebuilding a Solar Powered Website

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